Ageing murine female oligodendrocytes and male microglia transcriptomically converge with major depressive disorder in the prefrontal cortex

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April 9, 2026

Abstract

Major depressive disorder (MDD) is a highly morbid, and demographic-specific psychiatric disorder. Past studies have highlighted distinct sex and non-neuronal cell type signatures for major depressive disorder in the human prefrontal cortex. Age is not a well-studied variable in this context, but it has been indirectly linked to the molecular basis of MDD. We used the Allen Brain Cell Atlas to sex- and age-specifically define non-neuronal cell types in the mouse prefrontal cortex (PFC). We compared differentially expressed genes from these PFC-female-aged vs. PFC-female-adult and PFC-male-aged vs. PFC-male-adult cell types to human major depressive disorder-related genes, and identified overlapping biological pathways. Our findings identify pericytes are a novel cell type of interest, implicate myelination in female oligodendrocytes as a continued area of interest, and highlight interleukin signaling in male microglia as a result that validates existing literature at the intersection of MDD and the ageing PFC. Our study demonstrates the value of region-, cell type-, sex-, and age-specific transcriptomics and major depressive disorder characterization.

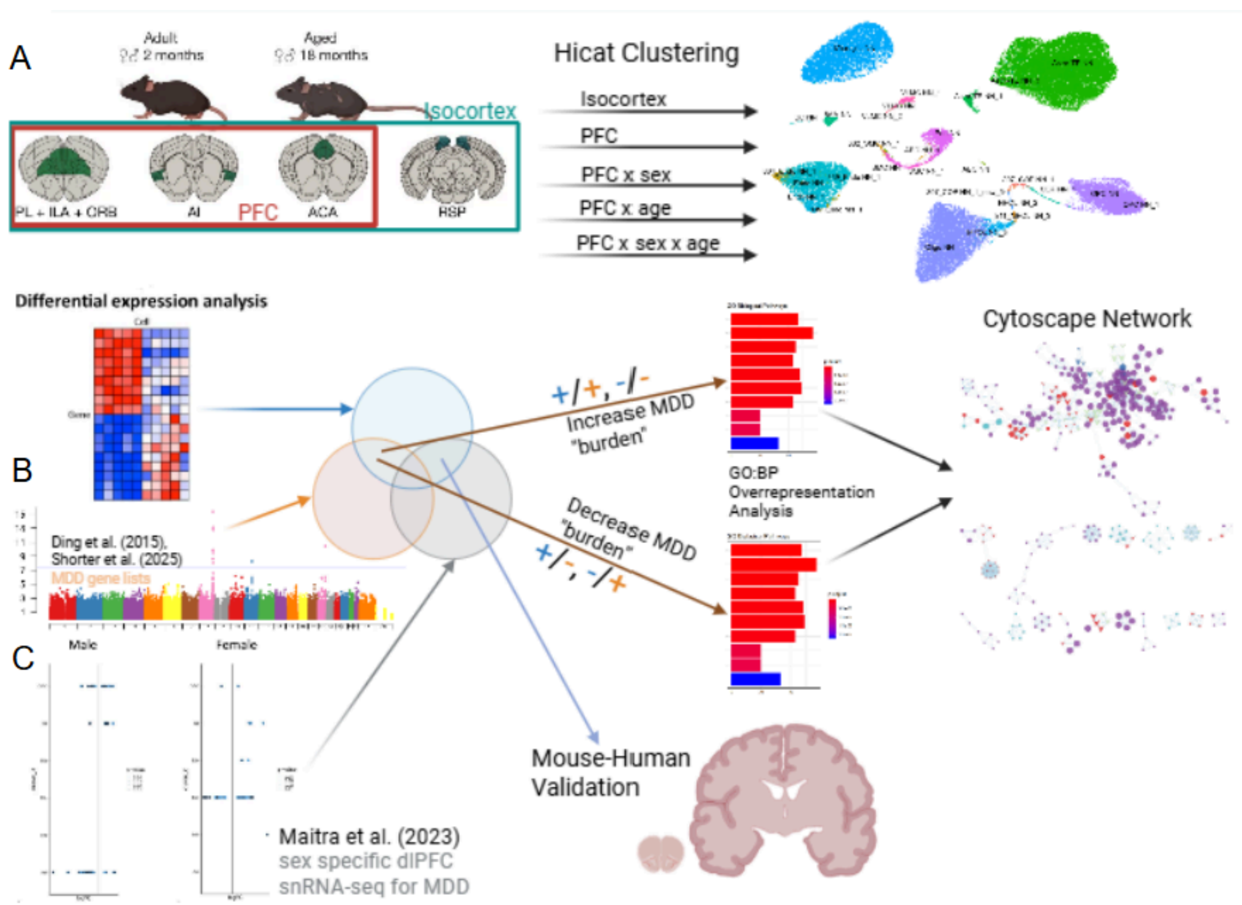

Figure 1. Graphical abstract. A) Subsetting and clustering of Allen Brain Cell Atlas. B) Differentially expressed gene overlap with major depressive disorder gene lists and pathway enrichment. C) Differentially expressed gene overlap with region-, sex-, and cell type-specific major depressive disorder gene lists.

Introduction

In 2021, depressive disorders were the second-highest contributor to morbidity in the Americas (WHO, 2022). MDD, specifically, has sex-specific presentations, with females having peak prevalence of MDD from the ages of 60-64, and males from 75-79. Furthermore, females are around twice as likely to receive an MDD diagnosis than males across their lifetimes (Li et al., 2023).

From a transcriptomic perspective, non-neuronal cells in the prefrontal cortex (PFC) are strongly implicated in the sex-specific biological aetiology of MDD. In humans, Maitra et al. (2023) performed a sex-stratified analysis of single nucleus RNA-seq (snRNA-seq) of the dorsolateral PFC (dlPFC) from MDD cases and controls, finding that all cell types presented sex-specific differentially expressed genes. Non-neuronal cells displayed the largest number of MDD differentially expressed genes and the greatest degree of sex specificity. Astrocytes and oligodendrocyte precursor cells had many MDD differentially expressed genes in males (astrocytes: 90, oligodendrocyte precursor cells (OPCs): 54) while the female counterparts had none. Conversely, female microglia had 74 differentially expressed genes while the male microglia had none (Maitra et al., 2023). With respect to mice, Paden et al. (2020) employed the four core genotypes design and found developmental hormone exposure and genetic sex contributed uniquely to depressive behaviours from bulk RNA-seq of the PFC. They found little overlap in gene expressions affected by chronic stress in males and females. Chronic stress in males appeared most strongly in the PFC, with genetic contributions highlighting granulocyte adhesion, which is a process motivated by astrocytes in the brain, and coagulation involving fibrinogen, which can promote OPC to astrocyte differentiation. For females, chronic stress appeared most strongly in the basolateral amygdala, with gonadal contributions highlighting transcription factors and the platelet glycoprotein VI aspect of coagulation. The inflammatory and cell type involvement of chronic stress in genetically male mice aligns with the male results from Maitra et al. The importance of different aspects of the same coagulation process in genetically male and gonadally female mice reflects the finding of similar pathway, but different cell-type and gene expression, dysregulation in males and females with MDD from Maitra et al. too. Overall, the gonadal and genetic specificity in gene expression for depressive behaviours, which reflects the sex-specificity of MDD displayed in Maitra et al., help validate the use of the mouse PFC to investigate human MDD transcriptomic signatures.

Age-specificity in the transcriptomic signatures of MDD in the PFC is poorly studied, and age-by-sex specificity even less so. Age has been related to MDD indirectly by Ding et al. (2015), who suggest that differentially expressed genes in MDD cases vs. controls in the brain have a strong overlap with age-related pathways over MDD-comorbid neuropsychiatric phenotypes. In a cross sectional approach, Shorter et al. (2025) found a handful of risk loci for early-onset and late-onset depression in a Nordic cohort. The early-onset and late-onset loci map to mutually exclusive gene sets, suggesting that the molecular basis of MDD is age-specific.

To address all PFC-specific, non-neuronal cell type-specific, sex-specific, and age-specific considerations for the transcriptomic basis of MDD, we leverage the high quality single cell RNA-seq (scRNA-seq) dataset from the Allen Brain Cell Atlas (Yao et al., 2023). This dataset provides us with age-, sex-, and region-annotated cells from over 100 mice, enabling age-, sex-, and region-aware cell cluster definition. Furthermore, we can then make transcriptomic comparisons between sex-, age-, and region-specifically defined cell types to evaluate the human-mouse conservation of the age-related biological signal for MDD in the PFC while respecting the uniqueness of male and female cells.

Methods

scRNA-seq source

Data from 32 285 gene transcripts by 1.2 million scRNA sequenced cells was analyzed from the Allen Brain Atlas (Jin et al., 2025; Yao et al., 2023). These cells were collected across male and female, adult (2 month old) and aged (18 month old) mice, and multiple regions of the brain.

The original dataset was filtered to only non-neuronal cells and then split into different region and demographic specific subsets (Supp. 1), which were each treated independently in the following analyses. The PFC was defined as the prelimbic-infralimbic-orbital (PL-ILA-ORB), anterior insula-claustrum (AI-CLA), and anterior cingulate area (ACA) regions (Chen et al., 2025). The AI is not usually considered as a part of the PFC, however the provided annotations did not separate the CLA from the AI. To retain as many cells as possible for stable clustering, the entire AI-CLA region was included.

Clustering

Clustering was performed using the scrattch.hicat R package (rather than scrattch.bigcat) as most subsets contained fewer than 100 000 cells. Default parameters were used in the iter_clust function, except that *sampleSize* was increased from 4 000 to 50 000 to improve cluster stability. Cluster purity was assessed using silhouette scores.

Each cluster was labelled with a cell type from the original dataset using the following heuristic:

- The clusters were sorted in descending order by number of cells
- Starting from the largest cluster, the cluster was given the most common subclass label (from metadata provided by Jin et al.) among the cells in the cluster
    - If this subclass label had already been assigned to a larger cluster then supertype label was considered
        - If this supertype label had already been assigned to a larger cluster, then cluster label was considered
            - If this cluster label had already been assigned to a larger cluster, then a new cluster name was created by appending "_new_cluster#" to the majority cluster label

This heuristic was used instead of the map_sampling function from the scrattch.hicat package because only one annotation scheme (subclass, supertype, or cluster label) could be used and just using subclass or supertype would not be granular enough (resulting in different clusters having the same name), but cluster label would be too granular (resulting in less interpretable cluster names). As well, map_sampling does not allow for novel cluster naming.

Differential expression analysis

Differential expression (DE) analyses were performed within each cluster based on sex, age, and region. DE analyses were also performed between clusters of the same assigned name for different subsets: PFC-female-aged vs. PFC-female-adult, PFC-male-aged vs. PFC-male-adult, PFC-female-aged vs. PFC-male-aged, and PFC-female-adult vs. PFC-male-adult comparisons.

DE analysis was performed using the Seurat R package (v. 4.3.0.1) (Hao et al., 2023). The FindMarkers function was used for female vs. male (sex), and aged vs. adult (age_cat) comparisons, and the FindAllMarkers function was used for multi-level region of interest (ROI) comparisons. Default parameters were used. Significant DE genes were defined as having a log2 fold change (log2FC) > 0.1 and Benjamini-Hochberg false discovery rate (FDR) < 0.05.

MDD gene list and DE gene overlap

To assess convergence of cell type- and region-specific sex and age DE with MDD, the overlap between DE gene lists with three MDD-associated gene sets were considered. All genes from these sets were mapped to mouse genes using the "Human and Mouse Homology Classes with Sequence information" file from the Mouse Gene Informatics database (Alliance of Genome Resources Consortium, 2024).

Table 1. Description of the 3 MDD gene lists used.

	1. Ding et al. (2015)	2. Shorter et al. (2025)	3. Maitra et al. (2023)
Study design	Meta-analysis of 8 human bulk RNA-seq studies (4 from the subgenual anterior cingulate cortex, 2 from the amygdala, and 2 from the dlPFC)	Genome wide association studies (GWASs) for general MDD diagnoses (mixed), MDD diagnoses before age 25 (early-onset or eoMDD), and MDD diagnosis after age 50 (late-onset or loMDD). Genome-wide significant single nucleotide polymorphisms (SNPs) were mapped to genes using FUMA's SNP2GENE	Sex stratified snRNA-seq study in human dlPFC between MDD cases and controls
Sample size	Male: 27 MDD, 26 control Female: 24 MDD, 24 control	Mixed: 151 582 MDD, 362 873 control Early-onset: 46 708 MDD, 106 824 control Late-onset: 37 168 MDD, 121 420 control	Male: 17 MDD, 16 control Female: 20 MDD, 18 control
Gene list size	566	Mixed MDD: 160 eoMDD: 17 (15 also identified in mixed	Male: Microglia: 0 Astrocyte: 188

		MDD) loMDD: 4 (all 4 identified in mixed MD)	Endothelial (matches to pericytes): 2 Oligodendrocyte: 16 Oligodendrocyte precursor cell: 64 Female: Microglia: 142 Astrocyte: 1 Endothelial (matches to pericytes): 0 Oligodendrocyte: 12 Oligodendrocyte precursor cell: 7
Number of mapped mouse gene homologs	505	Mixed MDD: 160 eoMDD: 16 loMDD: 4	Male: Microglia: 0 Astrocyte: 197 Endothelial (matches to pericytes): 2 Oligodendrocyte: 15 Oligodendrocyte precursor cell: 61 Female: Microglia: 120 Astrocyte: 0 Endothelial (matches to pericytes): 0 Oligodendrocyte: 9 Oligodendrocyte precursor cell: 5

The DE gene overlap between the Ding et al. and Shorter et al. was assessed for significance using hypergeometric tests. For each gene list, the universe size was set as the size of the union of all transcripts measured by Jin et al. and the mouse-mapped gene list (Supp. 2). The p-values were adjusted over all measurements using the FDR. Significant gene list overlaps were defined as having an FDR < 0.05.

Overlap with sex and cell type matched DE genes from Maitra et al. was only used to validate findings and qualitatively assess commonalities between mouse and human scRNA-seq analyses.

Functional enrichment

Enrichment for biologically relevant terms from the overlap of significant DE genes per cell type and the Ding et al. or Shorter et al. was performed separately using the gost function from the gprofiler2 R package (v. 0.2.4) (Kolberg et al., 2020). Default parameters were used, except *correction_method* was set to “fdr” and *evcodes* was set to TRUE. The September 1, 2025 Gene Ontology: Biological Processes (GO:BP) mouse annotation file (without Pathway Figure Optical Character Recognition and Inferred

Electronic Annotations) from the Bader lab was used as the reference set. Significant term enrichments were defined as having an FDR < 0.05. Only the DE genes from the PFC-female-aged vs. PFC-female-adult and PFC-male-aged vs. PFC-male-adult comparisons were queried for functional enrichment as they provided a sufficient number of DE and MDD related genes to analyse and the demographic specific cluster comparison best addressed our study objectives.

Per gene set, enrichment was performed in two directional batches. The ‘increase depressive burden’ batch had a DE gene log2FC sign that was concordant with the estimated effect size of the gene from the reference list, and the ‘decrease depressive burden’ batch had a DE gene log2FC sign that was opposite of the estimated effect size of the gene from the reference list (this is the “decrease depressive burden” batch).

Only the DE and MDD gene overlap from the largest telencephalonic astrocyte, microglia, pericyte, oligodendrocyte, committed oligodendrocyte precursor cell, and oligodendrocyte precursor cell clusters were considered. This is because they each had a sufficient number of overlap genes and presented as biologically relevant or novel.

Cytoscape network

The significant “increase depressive burden” and “decrease depressive burden” GO:BP term enrichments for the 6 chosen cell types were all imported into a Cytoscape EnrichmentMap (Shannon et al., 2003; Kucera et al., 2016). Each term was imported as a node that had annotations of cell type, increase or decrease burden, gene set term description, and the genes within the term that intersected with the associated DE and MDD gene overlaps. Parameter edge *similaritycutoff* was set to 0.25 and edge similarity coefficients were calculated using the Jaccard method.

The AutoAnnotate app was used to summarize the subgraphs into interpretable themes. Within the AutoAnnotate app, the MCL Cluster algorithm from the clusterMaker2 app was used to define the term clusters from each node’s similarity coefficients. As well, the WordCloud app algorithm that considered most frequent words within the cluster and adjacent words was used with parameters *max words per label* = 3, *minimum word occurrence* = 1, *adjacent word bonus* = 8 on the gene set description column to create cluster labels.

Only the DE gene overlap with the Ding et al. gene set was analysed in Cytoscape as this gene set provided the most enrichment terms and better matched the biological context of the transcriptomic data compared to the GWAS from Shorter et al.

Results

Non-neuronal scRNA-seq data within the isocortex was sex and age biased

Table 2. Number of non-neuronal cells and donors for the isocortex and PFC subsets.

Subset	Isocortex (# donors)	PFC (# donors)
female-aged	32 627 (5)	22 784 (3)
female-adult	9 696 (5)	5 567 (4)
male-aged	48 411 (9)	37 982 (6)
male-adult	30 960 (4)	24 517 (3)

Although Jin et al. collected cells from a balanced cohort of 108 mice, only 23 and 16 mice contributed to the non-neuronal cells in the isocortex and PFC, respectively. The overall pattern displayed, within region and sex categories, was a bias toward aged over adult cells, and, within region and age categories, more male than female cells. The result was the female-adult subset having far fewer cells than the others, despite having a comparable number of donors (Table 2), indicating that the original study's quality control and sampling procedures did not ensure sex and age balance within brain regions for non-neuronal cells. As well, the large number of donors that increases the overall validity of the data from Jin et al. did not contribute to all brain regions, and made the data subsets more vulnerable to individual noise.

Astrocytes were the most common cell identity within the PFC

Table 3. Number of cells in clustered subsets and their average silhouette scores.

	PFC-female-aged (silhouette score)	PFC-female-adult (silhouette score)	PFC-male-aged (silhouette score)	PFC-male-adult (silhouette score)
Telencephalonic Astrocyte (Astro-TE)	7828 (0.36)	2716 (0.84)	15899 (0.49)	7993 (0.48)
Microglia (Mic)	3622 (0.47)	726 (0.67)	5493 (-0.14)	4504 (0.55)
Oligodendrocyte (Oligo)	5044 (-0.04)	212 (0.24)	5351 (0.00)	2770 (-0.03)
Committed Oligodendrocyte Precursor (COP)	108 (0.27)	118 (0.39)	153 (0.25)	239 (0.32)
Oligodendrocyte	2948	1022	4747	2979

Precursor (OPC)	(0.72)	(0.81)	(-0.13)	(-0.01)
Pericyte (Peri)	619 (0.19)	95 (0.24)	931 (0.08)	737 (-0.04)
Endothelial Cell (Endo)	1311 (-0.12)	352 (0.21)	3379 (0.15)	2602 (-0.19)
Smooth Muscle Cell (SMC)	181 (0.15)	58 (0.27)	362 (-0.07)	270 (0.22)
Border Associated Macrophage (BAM)	115 (0.14)	76 (0.62)	231 (0.19)	250 (0.37)
Vascular Leptomeningeal Cells (VLMC)	136 (-0.14)	69 (-0.18)	170 (-0.09)	117 (0.20)
Olicogendrocyte Precursor, 1 (OPC NN_1)	0	0	252 (0.19)	214 (0.09)
Myelin Forming Oligodendrocytes (MFOL NN_3)	0	123 (0.01)	82 (0.22)	444 (0.25)
Telencephalonic Astrocyte, 1 (Astro-TE NN_1)	140 (0.10)	0	131 (0.07)	262 (0.14)
Telencephalonic Astrocyte, 3 (Astro-TE NN_3)	159 (0.01)	0	117 (0.08)	145 (-0.06)

In Table 3, compared to the other subsets, PFC-female-adult had a much smaller number of cells in its microglia and oligodendrocyte clusters.

Since the male subsets were larger, they were able to detect more granular cell subtypes not included in Table 3, such as other Endo, SMC, VLMC, and COP clusters, mature oligodendrocytes, and age-associated B-cells in the PFC-male-aged subset specifically. All of these cell subtypes were also detected in the whole isocortex, suggesting that at the scale of this study, the number of cells limits cluster granularity more than demographic heterogeneity.

The mean silhouette scores for all of the clusters per subset were quite low. However, these scores were comparable to those of the subclass level groupings from Jin et al. (Supp. 3). This poor cluster purity suggests that PFC non-neuronal cells harbour substantial granular heterogeneity that could not be robustly described with the hicat clustering algorithm at this sample size. The silhouette score metric is also problematic for scRNA-seq clustering as it favours compact, spherical clusters, which is not a useful bias in this context (Rautenstrauch & Ohler, 2025).

The PFC contained more sex DE genes than the isocortex

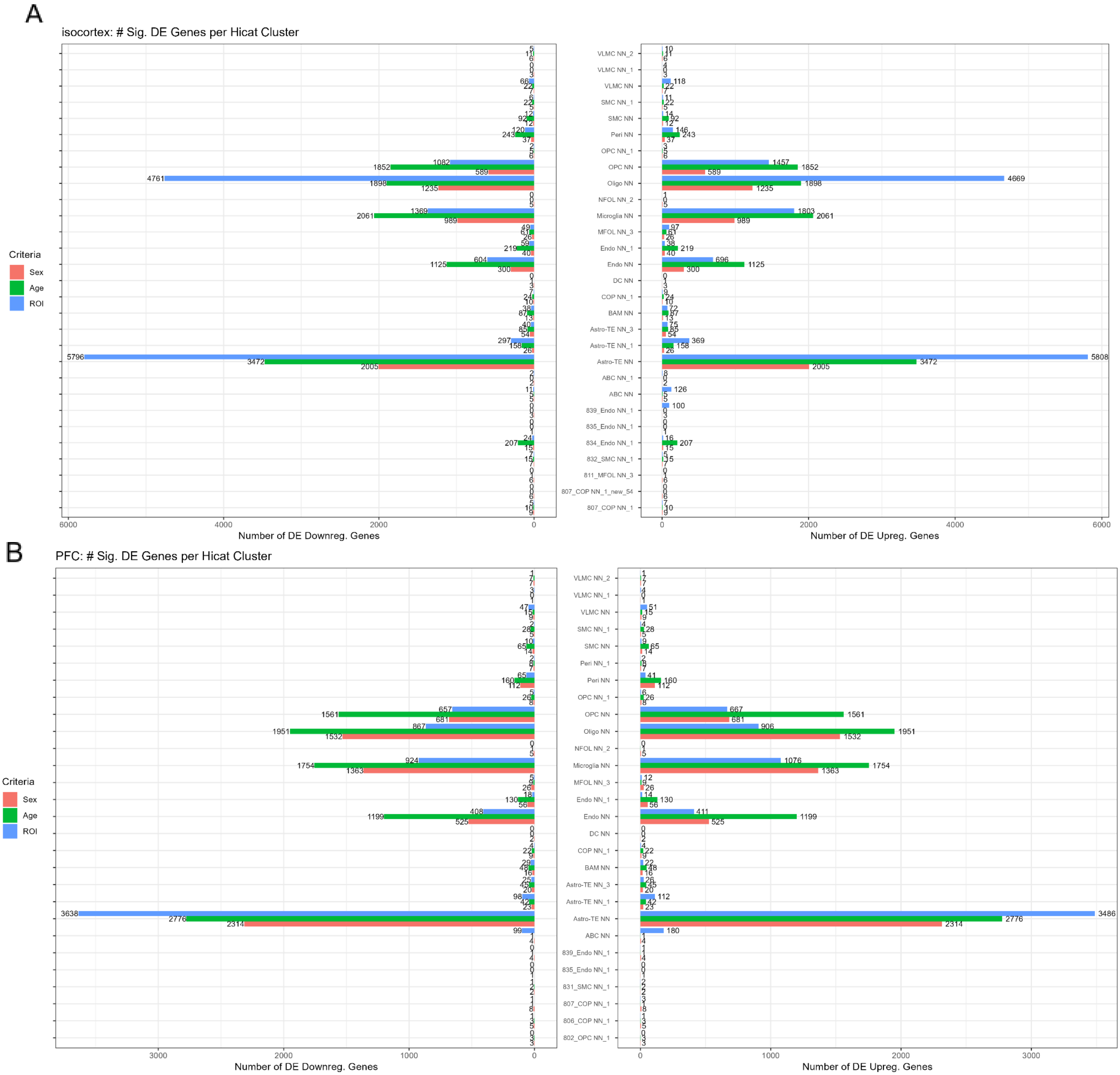

Figure 2. A) Number of significant DE genes in the whole isocortex within each identified cell type cluster by sex, age, and subregion (ROI). OPCs, pericytes, microglia, and endothelial cells have more age

DE genes than either ROI or sex. B) Number of significant DE genes in the whole PFC within each identified cell type cluster by sex, age, and ROI.

Figure 2A shows that the whole isocortex contained a large amount of subregional heterogeneity, with the number of significant DE genes per cell type being the largest or second largest per cell type compared to sex and age. Comparing Figure 2A to 2B, the whole PFC contained more sex and age than subregional heterogeneity in most cell types, except for astrocytes, age-associated B cells (ABC), and VLMC. Notably, subsetting the isocortex to just the PFC produced more sex-based DE genes in all cell types, except for the two smaller astrocyte clusters.

In both the isocortex and PFC, the cell types with the largest number of DE genes followed the order of: astrocytes, oligodendrocytes, microglia, OPCs, and endothelial cells (from most to least). In the isocortex, the second largest astrocyte cluster (Astro-TE NN_1) had more DE genes than pericytes, but the opposite was true in the PFC. In Figure 2A, there were also a few unique cell types that were detectable in only the isocortex, and not the PFC (myelin-forming oligodendrocytes and a new COP subtype that was not named in the original dataset). The PFC also had a few unique cell types that were a SMC subtype and two COP subtypes. The presence of cell types in the PFC and sex DE genes that were not detected in the isocortex highlighted that there may be PFC-unique signals lost by considering the whole isocortex.

Male oligodendrocytes had DE gene overlap with late-onset MDD in the ageing PFC

Table 3A. DE genes that overlap with the Shorter et al. eoMDD gene list. Genes above “/” increase and genes below “/” decrease depressive burden. Legend shows highlighted gene associations.

	Shorter2025 eoMDD	
Cell Type	PFC_male_aged vs PFC_male_adult	PFC_female_aged vs PFC_female_adult
Microglia	-------------------------------------	/ **Sdk1**
Oligo	Sorcs3 /	---
COP	Sox5 /	---
OPC	Zcchc7 / **Sdk1**	---
Astro-TE	Fibp / **Sdk1**,Sorcs3	/ Mmab,**Sdk1**,Sox5
Peri	-------------------------------------	---

Legend
Neuron
Nervous system disorder
Immune

Table 3B. DE genes that overlap with the Shorter et al. eoMDD gene list.

	Shorter2025 loMDD	
Cell Type	PFC_male_aged vs PFC_male_adult	PFC_female_aged vs PFC_female_adult
Microglia	-------------------------------------	---
Oligo	Bsn /	---
COP	-------------------------------------	---
OPC	-------------------------------------	---
Astro-TE	-------------------------------------	---
Peri	-------------------------------------	---

Due to the small number of eoMDD- and loMDD-associated genes from Shorter et al., very few subsets and cell types have DE gene overlaps with these MDD subtype-specific gene lists. However, the presence of any overlaps is then notable.

In the healthy ageing female PFC, the only overlaps occurred in microglia and astrocytes, which have DE genes that decrease early-onset depressive burden. In the healthy ageing male PFC, oligodendrocytes had overlaps that increase depressive burden for eo- and loMDD. *Sdk1* appeared as a down-regulated gene in ageing male OPCs and astrocytes, and ageing female microglia and astrocytes. Due to *Sdk1* having a positive estimated effect size of eoMDD from Shorter et al., *Sdk1* was then classified as decreasing depressive burden. The only other genes that appeared more than once were *Sorcs3* (in ageing male oligodendrocytes and astrocytes) and *Sox5* (in ageing male COPs and ageing female astrocytes), but they had inconsistent directionality between cell types and sexes. The one intersection with the loMDD gene list, *Bsn*, was in ageing male oligodendrocytes.

The ageing PFC contained more MDD related DE genes than the isocortex for certain cell types

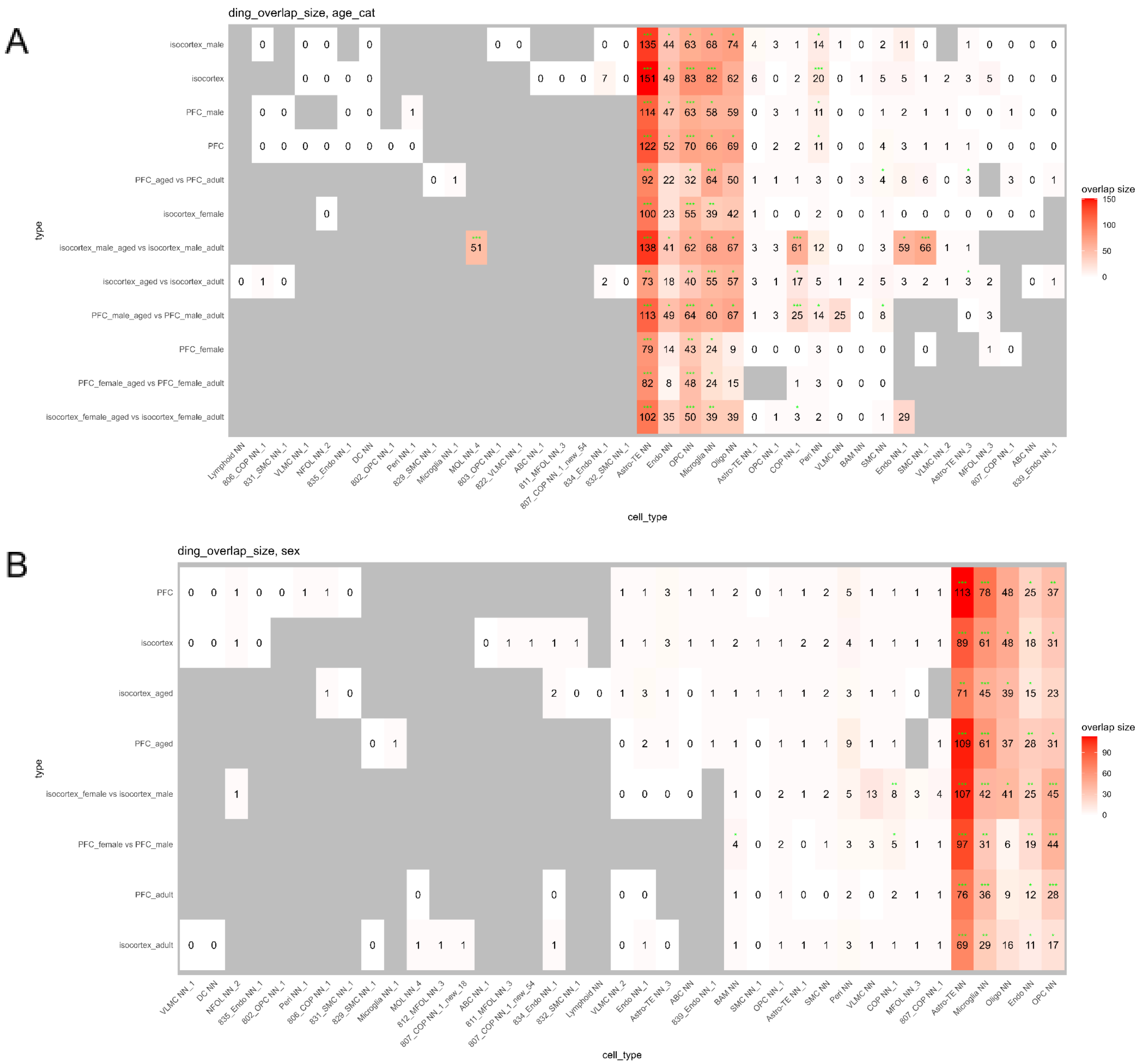

Figure 3. A) The number of age based DE genes that overlap with the gene list from Ding et al. in several isocortex and PFC subsets across all cell type clusters detected in the subset. B) The number of sex based DE genes that overlap with the gene list from Ding et al. in several isocortex and PFC subsets across all cell type clusters detected in the subset. A & B) The rows and columns are hierarchically ordered. The grey tiles mean that this cell type was not identified in this subset. Green stars indicate the statistical significance of the MDD gene list-DE gene overlap, assessed by hypergeometric tests with FDR adjustment. * = FDR < 0.05, ** = FDR < 0.001, *** = FDR < 0.0001.

In Figure 3A, the PFC had more age-based DE gene overlaps with the MDD-related gene list from Ding et al. than the isocortex in oligodendrocytes and endothelial cells. PFC-aged vs. PFC-adult had more overlap than isocortex-aged vs. isocortex-adult in astrocytes cells and microglia. PFC-male-aged vs. PFC-male-adult had more overlap than isocortex-male-aged vs. isocortex-male-adult in endothelial, OPC, pericyte, and VLMC cells.

In Figure 3B, the PFC, PFC-aged, and PFC-adult had more sex-based DE gene overlaps with the MDD-related gene list from Ding et al. than the isocortex, isocortex-aged, and isocortex-adult, respectively, in astrocytes, microglia, endothelial, OPC, and pericyte cells. The PFC-female-aged vs. PFC-male-aged and PFC-female-adult vs. PFC-male-adult cell type specific comparisons had no DE genes and, therefore, did not appear in Figure 3B. The mature oligodendrocyte cell type appeared in only the isocortex-male-aged vs. isocortex-male-adult comparison as this cell type was not detectable at the PFC level and isocortex-female level, possibly due to low cell numbers. This cell type had a strong overlap with Ding et al. (51) and many significant male, age DE genes (1022) despite having only 167 isocortex-male-aged and 378 isocortex-male-adult cells (compared to the similar cell type, oligodendrocytes, with an overlap of 67 genes, 1977 DE genes, 7343 isocortex-male-aged and 3989 isocortex-male-adult cells).

In Figure 3B, the hierarchically clustered rows matched PFC and isocortex subsets (i.e. whole PFC and isocortex, isocortex-aged and PFC-aged, isocortex-female vs. isocortex-male and PFC-female vs. PFC-male, and PFC-adult and isocortex-adult were next to each other). This highlighted that the MDD-related gene overlap with sex based DE genes within the isocortex may have been sufficiently captured by the PFC alone.

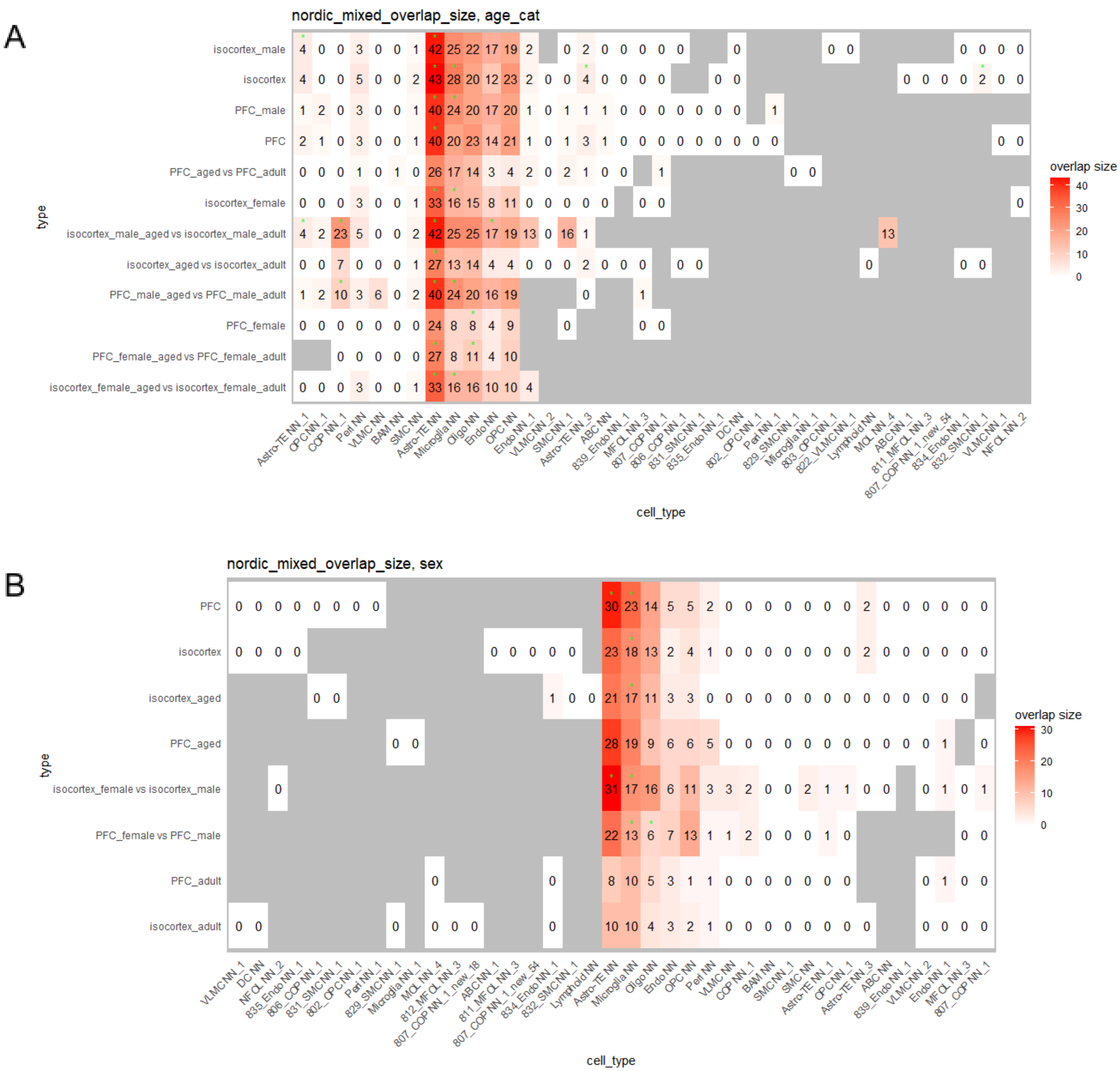

Figure 4. A) The number of age based DE genes that overlap with the mixed gene list from Shorter et al. in several isocortex and PFC subsets across all cell type clusters detected in the subset. B) The number of sex based DE genes that overlap with the mixed gene list from Shorter et al. in several isocortex and PFC subsets across all cell type clusters detected in the subset. A & B) The rows and columns are hierarchically ordered. The grey tiles mean that this cell type was not identified in this subset. Green stars indicate the statistical significance of the MDD gene list-DE gene overlap, assessed by hypergeometric tests with FDR adjustment. * = FDR < 0.05, ** = FDR < 0.001, *** = FDR < 0.0001.

Similar to the overlap with Ding et al., the PFC subsets captured a comparable number of sex and age DE gene overlaps with the mixed-MDD gene list from Shorter et al. as their corresponding isocortex subsets. Astrocytes, microglia, oligodendrocytes, OPC, and COP cells stood out as having the largest overlaps. In Figure 4A, the mature oligodendrocyte cell type in the isocortex-male-aged vs. isocortex-male-adult comparison also had a notable number of overlaps with Shorter et al., but it was not significant. As well, the smooth muscle cell type in the isocortex-male-aged vs. isocortex-male-adult comparison had more overlaps with Shorter et al. than Ding et al. In general, the DE gene overlaps with Shorter et al. were smaller and less significant than the overlaps with Ding et al. This was likely due to the GWAS-based analysis of Shorter et al. only capturing the static genetic component of MDD, which does not account for even tissue level specificity.

Based on Figures 3 and 4, the scope of the study was narrowed to only the age DE gene overlaps with Ding et al. in astrocytes (Astro-TE NN), microglia, oligodendrocytes, OPC, COP, and pericytes from the PFC-female-aged vs. PFC-female-adult and PFC-male-aged vs. PFC-male-adult subset comparisons. The Ding et al. gene set was chosen because it best matched the transcriptomic and PFC focus of this study. The six cell types were chosen because they align with current literature about cell type-specific contributions to MDD in the PFC (astrocytes, microglia, oligodendrocytes, OPC) (Maitra et al., 2023) or represent novel, more granular cell types (COP, pericytes). The two subset comparisons were selected because comparing the same cell types, but defined by independent clusterings at a region-, sex-, and age-specific level, most directly accounts for the region-, sex-, and age-specificity of MDD.

Circulatory and interleukin terms highlighted in MDD-related DE genes in male ageing PFC

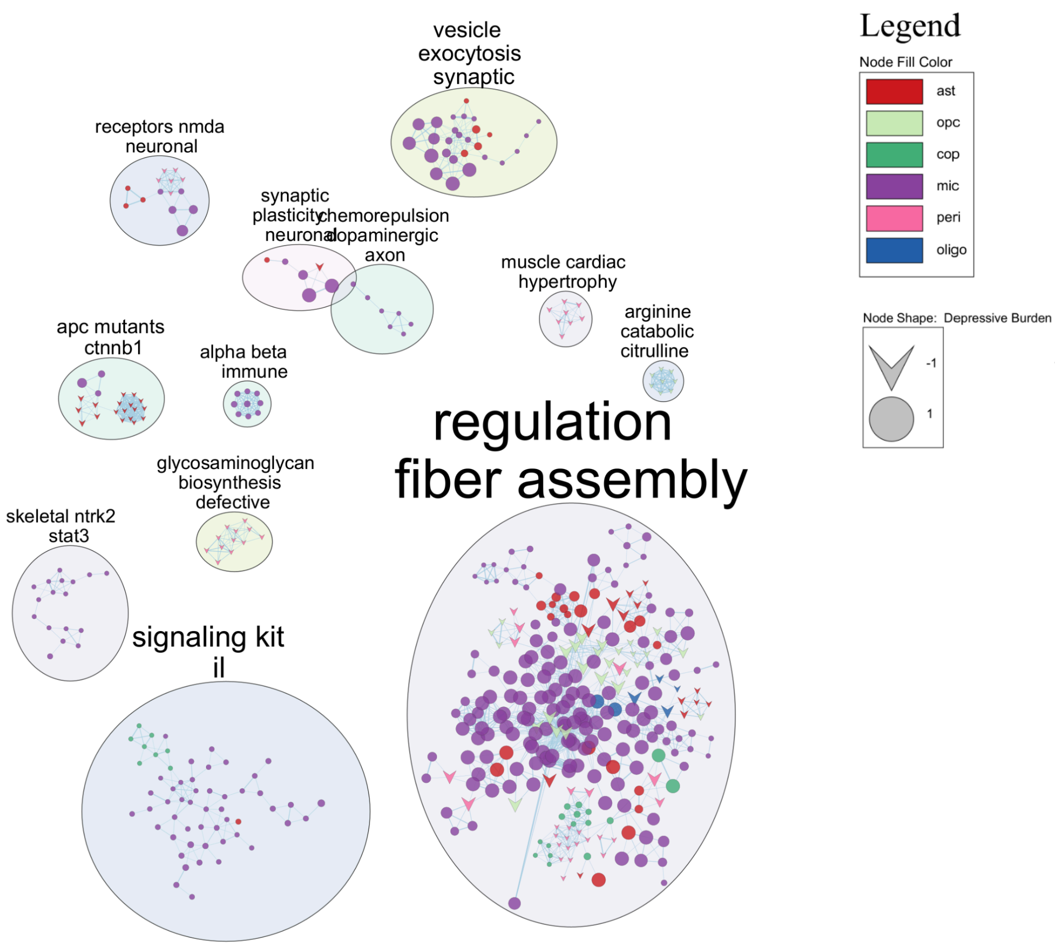

Figure 5. Cytoscape summarized significant GO:BP term enrichments in PFC-male-aged vs. PFC-male-adult from the decrease depressive burden and increase depressive burden DE gene overlaps with the MDD-related gene list from Ding et al. in astrocytes (ast), microglia (mic), oligodendrocytes (oligo), OPC (opc), COP (cop), and pericytes (peri). -1 = decrease depressive burden, 1 = increase depressive burden.

In the ageing male PFC, the cardiac cluster in Figure 5 was composed of only pericytes and the terms are related to negative regulation of heart growth. This may reflect reduced pericyte contractility and improved blood flow to the PFC. The interleukin (IL) signaling cluster connected angiogenesis to VEGFR to T cell activation to several tyrosine kinases in microglia, as well as a COP connection with respect to neuronal damage. All of the nodes in this cluster came from the increasing depressive burden enrichments, suggesting that inflammatory immune activation is present in MDD. The fiber assembly cluster related high heart rate, vesicular transport, and insulin response, pointing towards a catabolic state in MDD-related ageing DEs in the male mouse PFC. The strong presence of microglia representation in the IL, NTRK2/STAT3, and alpha-beta immune clusters (all related to inflammation) helped validate these GO:BP enrichment results.

Table 4. Number of significant GO:BP term enrichments in PFC-male-aged vs. PFC-male-adult from the decrease depressive burden and increase depressive burden DE gene overlaps with the MDD-related gene list from Ding et al.

Cell type	Decrease burden	Increase burden
Astro-TE	138	278
Microglia	0	544
Peri	184	0
Oligo	23	19
COP	11	63
OPC	107	0

At the broader cell type-level, of the 968 term nodes imported into Cytoscape, 544 were from microglia (Table 4). As well, microglia only had significant terms in the increase burden direction, suggesting that ageing microglia in the male PFC contribute to the molecular convergence of MDD and ageing. Conversely, pericytes and OPCs only had significant terms in the decrease burden direction, suggesting that ageing pericytes and OPCs in the male PFC may be more strongly dysregulated in the context of MDD.

GABA degradation terms highlighted in MDD-related DE genes in female ageing PFC

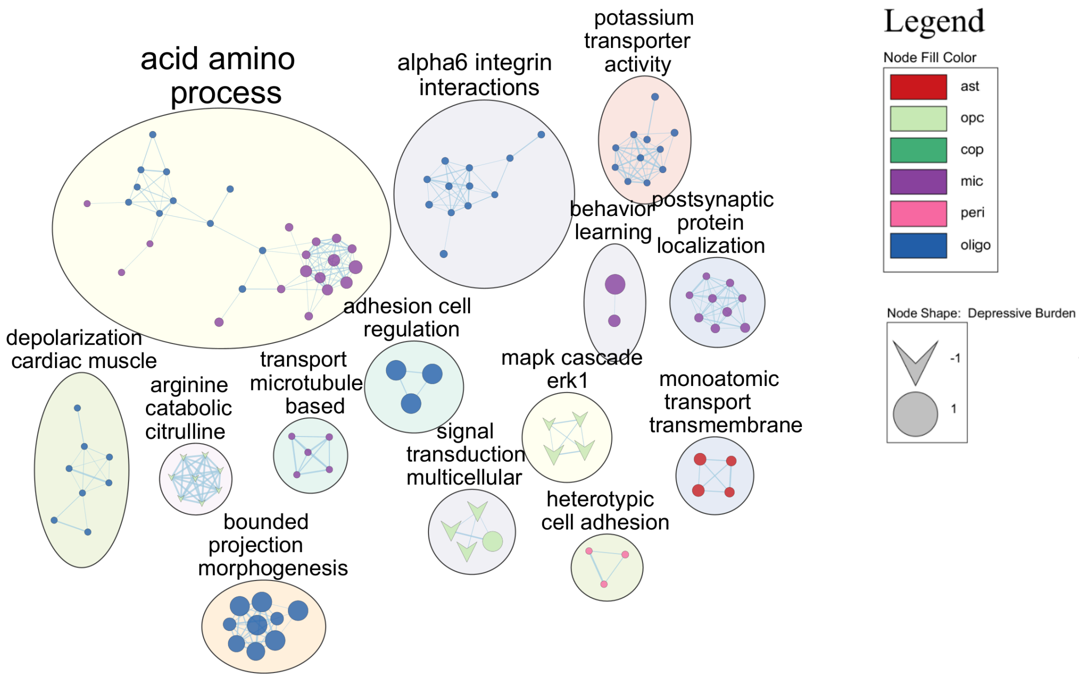

Figure 6. Cytoscape summarized significant GO:BP term enrichments in PFC-female-aged vs. PFC-female-adult from the decrease depressive burden and increase depressive burden DE gene overlaps with the MDD-related gene list from Ding et al. in astrocytes (ast), microglia (mic), oligodendrocytes (oligo), OPC (opc), COP (cop), and pericytes (peri). -1 = decrease depressive burden, 1 = increase depressive burden.

In the ageing female PFC, the amino acid process cluster in Figure 6 related to the GABA shunt in oligodendrocytes, which is catabolic in nature, and matched the catabolic trend in the male pathway summary. This cluster works with the current literature on decreased GABA in the PFC contributing to MDD and inhibited myelination (Sacchet & Gotlib, 2017; Hamilton et al., 2016). Conversely, the OPC MAPK/ERK cascade cluster may reduce myelination, while being classified as decreasing depressive burden (Serrano-Regal et al., 2020). The heterotypic cell adhesion pericyte cluster has terms related to negative regulation of this process, indicating high vascular permeability is associated with increased depressive burden.

The arginine to citrulline cluster was consistent in burden direction (decreases) and cell type composition (OPCs only) in males and females. The cluster terms related to the negative regulation of arginine to citrulline in OPCs. Astrocyte-related terms, which increase depressive burden in both sexes, refer to positive regulation of monoatomic cation (in females) and more specifically potassium, calcium, and calmodulin kinase (in males).

Table 5. Number of significant GO:BP term enrichments in PFC-female-aged vs. PFC-female-adult from the decrease depressive burden and increase depressive burden DE gene overlaps with the MDD-related gene list from Ding et al.

Cell type	Decrease burden	Increase burden
Astro-TE	4	21
Microglia	0	79
Peri	3	11
Oligo	8	107
COP	0	0
OPC	47	24

Of the 251 term nodes imported into Cytoscape, 115 were from oligodendrocytes (Table 5). Similar to the male term enrichments, microglia only had significant terms in the increase burden direction and OPCs had mostly terms in the decrease burden direction. Female pericytes also had more increase than decrease burden terms, and COPs had no enrichments because there are no DE gene overlaps with Ding et al. Overall, there were almost 4 times fewer significant female than male term enrichments, highlighting the lack of sex specificity in the MDD gene list from Ding et al. and in the Gene Ontology as a whole.

Sex-, region-, and cell type-specific human-mouse MDD-related DE gene overlap highlighted general neuropsychiatric load

Table 6A. DE genes that overlap with the matched cell type-specific Maitra et al. MDD gene lists. Legend shows highlighted gene associations.

	Maitra2023	
Cell Type	PFC_male_aged vs PFC_male_adult	PFC_female_aged vs PFC_female_adult
Mic	--	Meg3,Ccl4,Bcl2a1b,Socs3,**Cntnap2**,**Rbfox1**,**Synpr**,**Nrg**
Oligo	**Tenm2**,Etl4,Hspa1a,Lrmda	**Zfp804b**
COP	**Nrxn3**	---
OPC	**Gpr17**,**Sema5a**,**Ryr2**,**Sema3e**,**Cntnap4**,Tuba1a,Aldoc,**Kcnh7**,Actb,Marcks,Mgat4c,**Robo2**,**Gabrg3**,**Cnr1**,Atp1b2,Prdx1,**Frmpd4**,**Dlgap2**,Ddit4,Itga8,Nkain2,Marcksl1,**Lingo2**,**Nrxn3**,Mt2,**Grin2a**	Mbp
Astro	Glul,Apoe,Htra1,Aqp4,Atp1a2,Gabra2,Mt1,Slc39a12,Cst3,Ncan,Ptn,Plpp3,Gpr37l1,Myorg,Rhob,Fam107a,Hes1,Zfp36,Bcl6,Fads2,Ctsd,Gjb6,Ednrb,Cspg5,Grin2c,Dag1,Tfrc,Ksr2,Galntl6,Socs3,Tubb4a,Emx1,Tril,S1pr1,Lrrc8a,Kcnj10,Slc6a11,Ranbp3l,Cadm4	---
Peri	--	---

Legend
Neuron
Nervous system disorder
Immune
Cell type marker

Table 6B. DE genes that overlap with the matched cell type-specific Maitra et al. and Ding et al. MDD gene list.

	Ding2015, Maitra2023	
Cell Type	PFC_male_aged vs PFC_male_adult	PFC_female_aged vs PFC_female_adult
Mic	--	--
Oligo	--	--
COP	**Nrxn3**	--
OPC	**Nrxn3**,**Ryr2**	Mbp
Astro	Bcl6,Fads2,Rhob	--
Peri	--	--

In Table 6A, OPCs and astrocytes in males overlapped strongly with the same cell types in the male MDD DE genes from Maitra et al. (26 and 40 genes, respectively), while microglia showed better overlap in females (8 genes). Ageing female microglia and ageing male astrocytes were to only cell types that had immune related DE gene overlaps with Maitra et al., which helped validate our results. In Table 6B, the intersection of all three gene sets, this study's age specific DE genes, Ding et al., and Maitra et al., highlighted that males still had a larger overlap than females.

Cell type marker genes appeared in the intersection of DE genes from male ageing OPCs and astrocytes and Maitra et al., and the OPC marker *Mbp* appeared in the intersection of DE genes from female ageing OPCs, Maitra et al., and Ding et al. The appearance of cell type markers in these overlaps suggested that there may be dysregulation of cell type-characteristic behaviours in the convergence of MDD and ageing. In all other cell types, there were few to no overlaps with Maitra et al. despite the paper matching the sex, cell type, and brain region of this study, which left mouse vs. human (mice do not have a well defined homologue to the primate dlPFC) and ageing vs. MDD as key differences. There were no genes at the intersection of this study's age specific DE genes, Ding et al., Maitra et al., and any of the mixed MDD, eoMDD, or loMDD gene lists from Shorter et al.

Discussion

Our findings support the importance of the PFC in MDD and sex- and cell type-specificity in the convergence of ageing in mice and MDD. We highlight similar themes in pathway enrichments of male and female ageing murine PFC DE gene overlaps with human MDD DE genes, but different mechanisms. The female heterotypic cell adhesion pericyte GO:BP term cluster relates to negative regulation of this process, which suggests high vascular permeability, which is often related to the immune response, increases depressive burden. However, immune activation increasing depressive burden is more directly detected in males by microglial IL signaling, T-cell activation, and COP inflammatory responses to neuronal damage. In females, we relate a catabolic state to GABA degradation in oligodendrocytes. Current literature connects decreased GABA to myelination inhibition, suggesting decreased myelination in the PFC may increase depressive burden (Sacchet & Gotlib, 2017; Hamilton et al., 2016). Conflictingly, the female OPC MAPK/ERK cascade cluster may oppose this myelination hypothesis.

Activation of this pathway increases OPC population but inhibits differentiation into oligodendrocytes, reducing myelination, while being classified as decreasing depressive burden (Serrano-Regal et al., 2020). This inconsistency may be due to a species specific difference or the lack of sex-specificity of the Ding et al. MDD gene list. In males, we relate a catabolic state to microglial insulin receptor activation, and therefore glycolysis, at the intersection of MDD and ageing.

In females, we connect the general process of monoatomic cation transport in ageing PFC astrocytes to increased MDD burden, whereas in males, we find the similar, more specific potassium, calcium, and calmodulin kinase channel terms may increase MDD burden. The astrocytic calmodulin kinase pathway is commonly studied as a downstream effect of neuronal and glial potassium release in relation to spinal injury, where it promotes plasticity in astroglia networks. This remodelling may also spread inflammatory signals beyond the area of injury. As well, in the medial PFC, calmodulin kinase activation has a negative effect on working memory which is often impaired in MDD (Runyan et al., 2005). Furthermore, broader terms being in the female and more specific terms being in the male GO:BP enrichment suggest that there may be sex-specific subpathways in calmodulin kinase activation within the ageing PFC-MDD overlap context. The only shared cluster in pathway enrichments in both sexes is the negative regulation of arginine to citrulline in OPCs. This cluster's terms relate to the negative regulation of arginine to citrulline in OPC which further suggests lower Neutrophil Extracellular Trap production and less immune activation decreases depressive burden (Maronek & Gardlik, 2022).

These pathway and exact cell type identifications are highly tentative as our DE gene results from PFC-female-aged vs. PFC-female-adult and PFC-male-aged vs. PFC-male-adult comparisons show modest alignment with region-, sex-, and cell type-matched MDD vs. control DE gene lists. The genes that intersect at the ageing mouse and human MDD datasets (matched by region, sex, and cell type) tend to be associated not with mood regulation, but with other neuropsychiatric conditions including Alzheimer's disease, schizophrenia, and autism spectrum disorder. For instance, *Apoe, Htra1,* and *Hspa1a* are all neurodegeneration-related genes that are in the intersection of our male PFC age-DE genes, and the male cell type-specific MDD DE genes from Maitra et al. (Raulin et al., 2022; Hjæresen et al., 2026; Dong et al., 2022). Furthermore, when we also intersect the Maitra et al. overlaps with MDD DE gene meta-analysis from Ding et al., all of the genes that are common among these three sources have been linked to autism (Vaags et al., 2012; Soueid et al., 2016; Singh et al., 1993; Garbett et al., 2008; Sun et al., 2018), except for *Rhob*, which is related to cerebral palsy (Wu et al., 2024). Even at the least tissue-specific level, male oligodendrocyte PFC age DE overlap with the loMDD GWAS, *Bsn,* highlights an MDD comorbid disorder. In cases of G72-mediated schizophrenia mouse models, increased *Bsn* expression correlates with reduced *Sirt2* and decreased myelination, alluding to a role in the disruption of oligodendrocyte function (Filiou et al., 2022). *Bsn* also has a positive estimated effect size on loMDD from Shorter et al., suggesting that healthy ageing in male, PFC, murine oligodendrocytes captures some biological convergence of heritable loMDD and schizophrenic symptoms. In relation to the eoMDD gene list, in cell types where *Sdk1* was a significant age DE gene, it was universally categorized as decreasing depressive burden because it has a positive estimated effect size on eoMDD from Shorter et al. and is more lowly expressed in aged than adult cells. A previous study negatively links *Sdk1* with chronic social defeat stress (CSDS) vulnerability in the male mouse PFC, disagreeing with the GWAS effect direction but agreeing with the direction of the age DE (Touchant & Labonté, 2022). This suggests that *Sdk1* may be an example of species specificity in the overlap of murine ageing and depressive symptoms. As well, CSDS tends to be epigenetically rather than genomically studied, suggesting the contribution of *Sdk1* to CSDS may not be strongly heritable (Bondar et al., 2025). Overall, the eoMDD list (SNP-based heritability = 0.11) had more overlaps than the loMDD list (SNP-based heritability = 0.06) (Shorter et al., 2025). Again, this suggests that the intersection of ageing and MDD, or the generalization from mice to humans in the context of MDD, tends to highlight genes characteristic of heritable, general neuropsychiatric load, rather than genes that capture the nuances of MDD.

There are also several structural limitations (with tractable improvements) that prevent a cohesive characterization of the region-, sex-, and cell type-specific overlap between MDD and ageing. First, the GWAS-based gene lists from Shorter et al. had less significant and interpretable overlaps with our DE genes, possibly due to their cell type-naive risk locus to gene mapping strategy. There are now extensive non-neuronal, cell type-specific expression quantitative trail loci (eQTLs) that would allow us to map risk loci to the genes whose expression they regulate (Bryois et al., 2022). Although brain region, sex, and age would not be accounted for, this cell type-specific eQTL mapping would help bring the static genomic MDD association from Shorter et al. closer to the dynamics of our non-neuronal cells of interest. Second, the MDD gene list from Ding et al. is not sex specific, possibly resulting in the female DE genes having fewer overlaps and biological process enrichments. But it is possible to stratify the studies analysed by Ding et al. by sex and apply their meta-analysis pipeline to produce a sex-specific MDD gene list as four of the eight studies that they analysed were composed entirely of females. Third, the lack of female and adult, and subsequently female-adult, non-neuronal (and also neuronal) cells in the PFC limits our power to produce a comparable number of stable, granular clusters and detect significant DE genes across all sex by age combinations. As well, the number of microglia and oligodendrocytes in the PFC-female-adult subset is disproportionately low while also being cell types of interest in our analyses, raising the question of how much our results may reflect chance variation, but also suggesting that substantially richer findings could emerge with more balanced sampling of these key cell types. A possible contributor to this lack of female-adult cells may be from the mitochondrial content filtering step from the quality control performed by Jin et al. Young female mouse brains tend to be more metabolically active than young male brains, possibly leading to inflated mitochondrial content scores and excessive quality control drop out (Knouse et al., 2022; Long et al., 2026). The disparity in adult-female vs. other age by sex subsets cell numbers is less glaring in the hippocampus, but it is still the subset with the fewest cells. Ideally, for our demographic specific clustering approach, we would be able to access all of the raw scRNA-seq data from Jin et al. and perform quality control in a region, age, and sex specific manner.

Overall, in both sexes, we see pericytes as a novel cell type of interest, myelination in female oligodendrocytes as a continued area of interest, and IL signaling in male microglia as a validating result at the intersection of MDD and age in the PFC. The species mismatch and lack of sex-specificity of the three MDD gene lists pose validity issues with our analyses, but the presence of significant mouse age DE gene and MDD-related gene overlaps along with biologically relevant MDD pathway recapitulation from these overlaps support our approach to investigating the convergence of age and MDD. To further apply our approach, we would like to see the next landmark spatially resolved mouse brain scRNA-seq to be done in adult and aged four core genotypes mice in order to better understand the endocrine vs. chromosomal interaction between sex and age. We expect the non-neuronal cell type-, PFC-, and genotype-specific transcriptomic MDD-age overlaps to reveal MDD-related hormonal pathways, which are notably missing from our age DE analysis, when considering the results of Paden et al. (2020), and to possibly help human to mouse generalization.

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Supplements

Find key scripts at: https://github.com/mding010705/BCB430_ABC_PFC_MDD.git

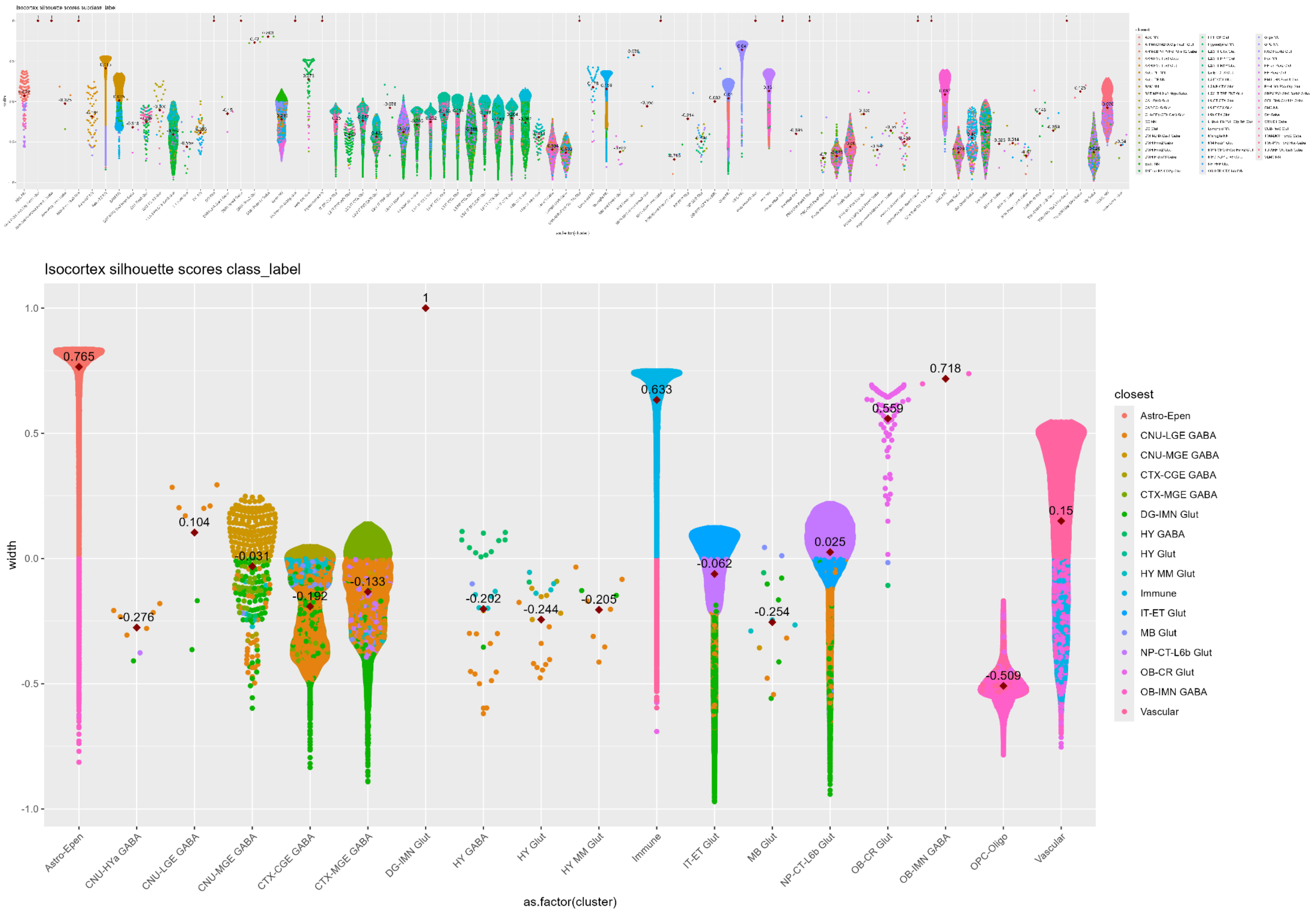
	isocortex	PFC
all	Clustered, DE analysis, Ding et al. and Shorter et al. mixed gene list overlap	Clustered, DE analysis, Ding et al. and Shorter et al. mixed gene list overlap
female	Clustered, DE analysis, Ding et al. and Shorter et al. mixed gene list overlap	Clustered, DE analysis, Ding et al. and Shorter et al. mixed gene list overlap
male	Clustered, DE analysis, Ding et al. and Shorter et al. mixed gene list overlap	Clustered, DE analysis, Ding et al. and Shorter et al. mixed gene list overlap
aged	Clustered, DE analysis, Ding et al. and Shorter et al. mixed gene list overlap	Clustered, DE analysis, Ding et al. and Shorter et al. mixed gene list overlap
adult	Clustered, DE analysis, Ding et al. and Shorter et al. mixed gene list overlap	Clustered, DE analysis, Ding et al. and Shorter et al. mixed gene list overlap
female-aged	Clustered, DE analysis compared to isocortex-female-adult and isocortex-male-aged, Ding et al. and Shorter et al. mixed gene list overlap	Clustered, DE analysis compared to PFC-female-adult (no DE genes compared to PFC-male-aged), Ding et al. and Shorter et al. mixed gene list overlap, Shorter et al. eoMDD and loMDD gene list overlap, GO:BP enrichment, Maitra et al. gene list overlap
female-adult	Clustered, DE analysis compared to isocortex-female-aged and isocortex-male-adult, Ding et al. and Shorter et al. mixed gene list overlap	Clustered, DE analysis compared to PFC-female-aged (no DE genes compared to PFC-male-adult), Ding et al. and Shorter et al. mixed gene list overlap, Shorter et al. eoMDD and loMDD gene list overlap, GO:BP enrichment, Maitra et al. gene list overlap
male-aged	Clustered, DE analysis compared to isocortex-male-adult and	Clustered, DE analysis compared to PFC-male-adult (no DE genes compared to

	isocortex-female-aged, Ding et al. and Shorter et al. mixed gene list overlap	PFC-female-aged), Ding et al. and Shorter et al. mixed gene list overlap, Shorter et al. eoMDD and loMDD gene list overlap, GO:BP enrichment, Maitra et al. gene list overlap
male-adult	Clustered, DE analysis compared to isocortex-male-aged and isocortex-female-adult, Ding et al. and Shorter et al. mixed gene list overlap	Clustered, DE analysis compared to PFC-male-aged (no DE genes compared to PFC-female-adult), Ding et al. and Shorter et al. mixed gene list overlap, Shorter et al. eoMDD and loMDD gene list overlap, GO:BP enrichment, Maitra et al. gene list overlap

Supplement 1. Table of all Allen Brain Cell Atlas cell subsets considered.

Gene list	Universe Size
Ding et al.	32 294
Shorter et al. mixed	32 285

Supplement 2. Table of universe sizes used for hypergeometric tests for significance of DE and MDD gene list overlap.



Supplement 3. Silhouette scores for the subtype (above) and class (below) level clusters from Jin et al. for the whole isocortex.